3 **(a)** **(i)** 5.00 cm **A1**

(ii) $\omega = \dfrac{2\pi}{T}$ **C1**

$= \dfrac{2\pi}{4.0} = 1.6 \text{ rad s}^{-1}$ **A1**

(iii) $v_0 = \omega x_0$ **C1**

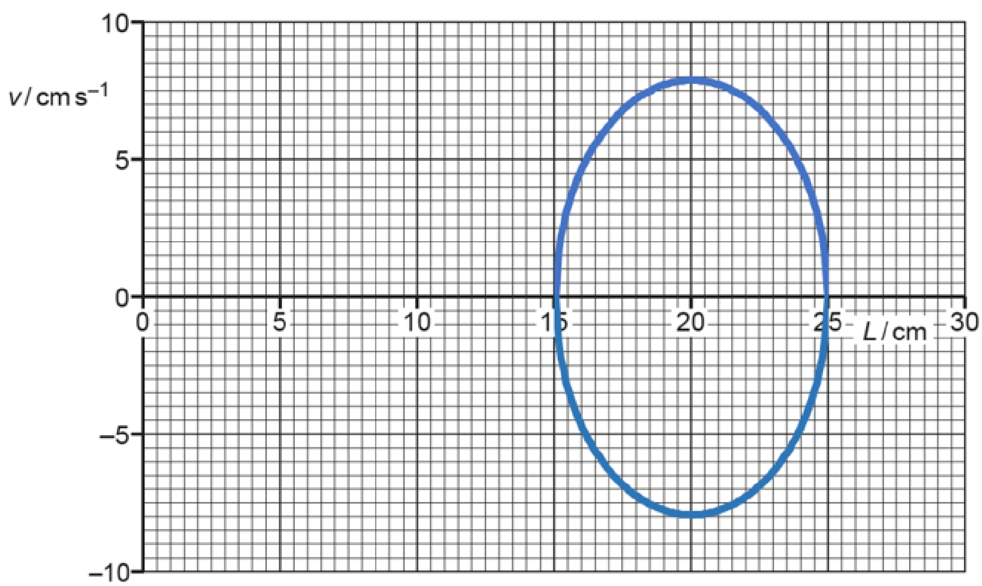
$= (1.6)(5.0) = 8.0 \text{ cm s}^{-1}$ **A1**

(b) Any three points below, 1 mark each **B3**

- initial pull was to the right
- distance from X to trolley (at equilibrium) is 20 cm
- initial motion undamped
- motion becomes damped at/from 12 s
- damping is light
- maximum speed at 1 s, 3 s, etc. / stationary at 2 s, 4 s, etc.

(c) sketch closed loop encircle (20,0) – see below **B1**

minimum L shown as 15 cm and maximum L shown as 25 cm **and**
minimum v shown as -8.0 cm s^{-1} and maximum v shown as 8.0 cm s^{-1} **B1**



4 **(a)** **(i)** Fundamental mode of vibration, $\lambda = 4L = 0.680$ or $L = 0.17$ m (not formed)